8.10 Differences between centrifugal pump and a reciprocating pump

Centrifugal pump	Reciprocating pump
Handles high volumes of liquid	Handles less volumes of liquid
Practically handles all types of liquid, oil, sewage, chemicals, paper pulp, jiggery, and liquids of high viscosity.	Used for pumping pure liquids and liquids with low viscosity.
Rotates at higher speeds	Moves at a slow speed
Not provided with suction valve and delivery valve	Provided with suction valve and delivery valve.
The discharge of liquid is continuous	The discharge is pulsating
Less maintenance cost	High maintenance cost
Less weight	High weight
Occupies less space	Occupies more space
Consists of few simple parts	Consists of complicated parts
The efficiency is low	The efficiency is high

8.11 Rotary pump

The types of rotary pumps are

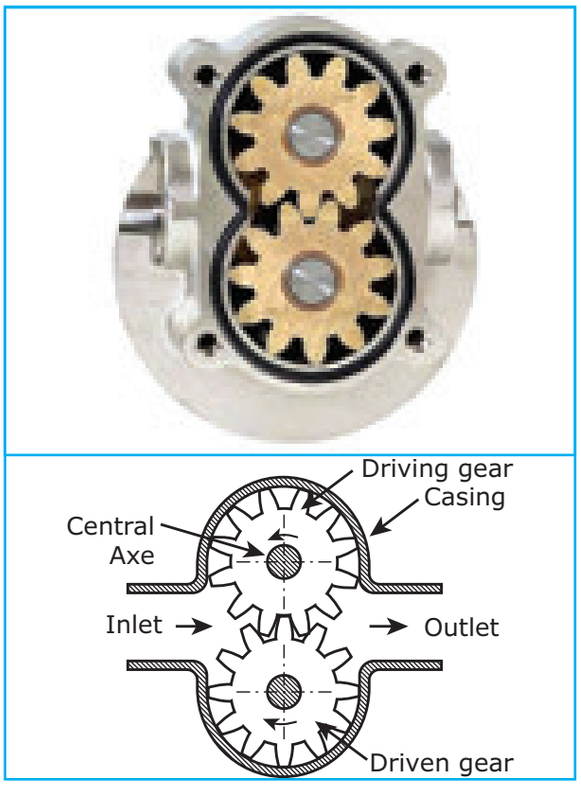
1. Gear pump
2. Vane pump
3. Lobe pump

Gear pump

In a gear pump, two gears are places inside a hollow casing in such a way that they rotate in opposite direction. One of these two gears is given rotary motion. The other gear will also rotate meshing with the first one. The shaft on the driving gear will be extending outside to get rotation from a prime mover. A little gap is left between the gears and the casing.

Working principle of gear pump

The inlet of the pump is immersed in the reservoir. The prime mover actuates the driving gear. This movement makes the other gear also to rotate. As these gears rotate, the liquid is trapped between their teeth and casing. The liquid is pumped out through the outlet of the pump. A gear pump is illustrated in figure.



Gear Pump

Uses

- Lubricating oil and liquids with high viscosity are pumped by pumps fitted with spur gears. Pumps with helical gears are useful in pumping water and chemicals. Gear pumps can handle pressures up to 140 kg/cm^2. The output can be 450 liters per minute.

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